

CYBERSECURITY INTERNSHIP

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University: Aditya University

Domain: Cybersecurity

Duration: 1st January 2026 to 30th April 2026

TASK-5: Malware Types & Behavior Analysis (Basic)

Malware (Malicious Software) refers to any program or code intentionally designed to cause harm to systems, networks, or users. Malware affects confidentiality, integrity, and availability of data and systems. The most common malware types include virus, worm, trojan, and ransomware, each having different characteristics and behaviors.

Virus:

A virus is a malicious program that attaches itself to legitimate files or programs. It requires user action, such as opening an infected file, to execute and spread. Once activated, a virus can corrupt files, slow down system performance, or destroy data. Viruses usually spread through infected email attachments, removable media like USB drives, or pirated software.

Worm:

A worm is a standalone malware that does not require user interaction to spread. Worms exploit network vulnerabilities and automatically propagate across systems. Because of their self-replicating nature, worms consume network bandwidth and system resources rapidly, often causing network congestion or system crashes.

Trojan Horse:

A trojan (Trojan Horse) disguises itself as a legitimate or useful application to trick users into installing it. Unlike viruses and worms, trojans do not self-replicate. Once installed, a trojan can create backdoors, steal sensitive information, log keystrokes, or give attackers remote access to the system.

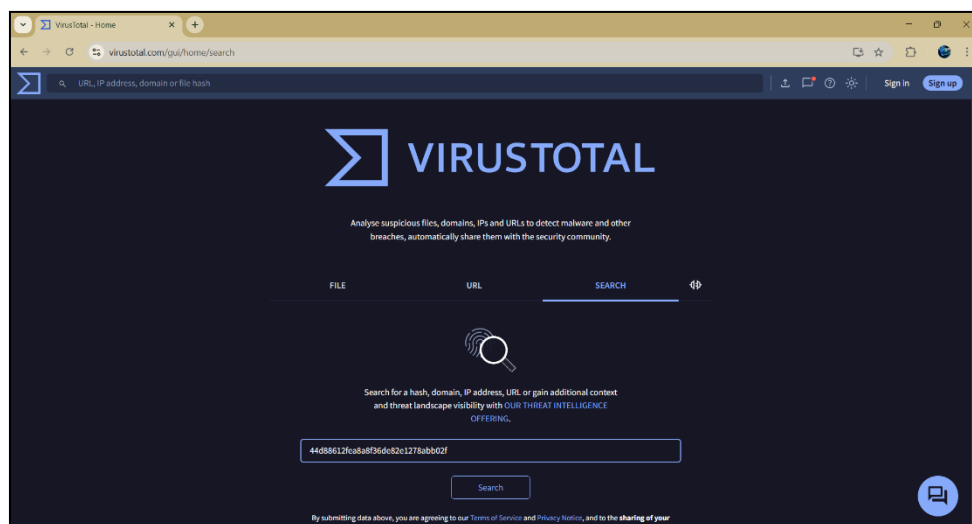
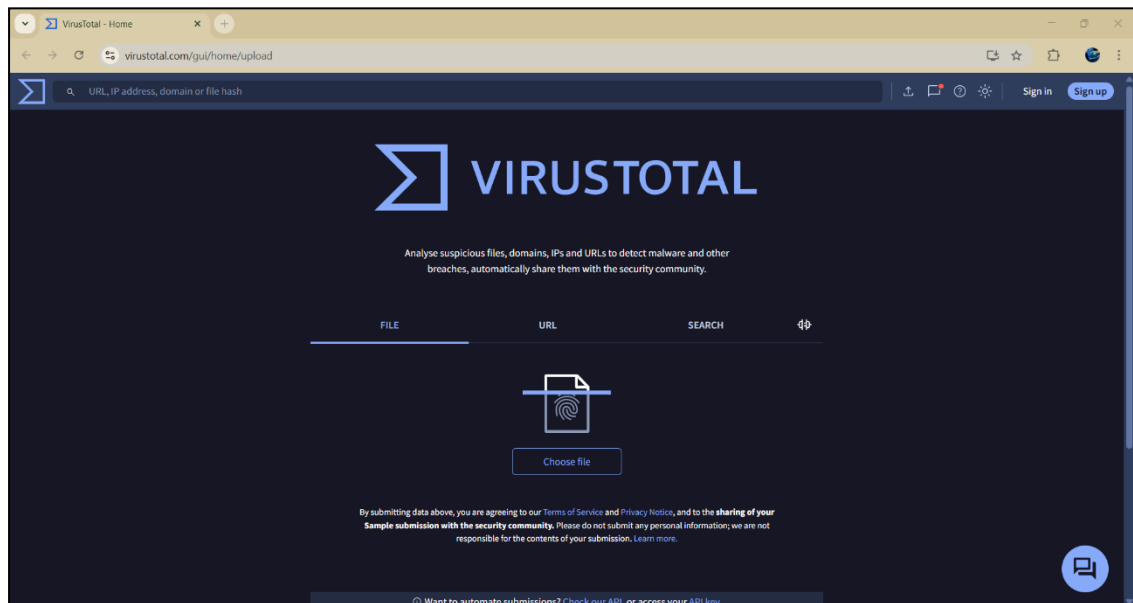
Ransomware:

Ransomware is one of the most dangerous forms of malware. It encrypts user data or locks the system and demands a ransom payment, usually in cryptocurrency, to restore access. Ransomware attacks can cause massive financial and operational damage to individuals, organizations, and governments.

- **Upload known malware samples (hashes) to VirusTotal:**

Directly uploading live malware files is unsafe and unethical. Therefore, analysis is done using **file hashes** (MD5, SHA-1, or SHA-256). A hash uniquely identifies a file without exposing the actual malware.

VirusTotal is a free online malware analysis service owned by Google. It scans files, URLs, IPs, and hashes using multiple antivirus engines. By submitting a malware hash, we can view detection results, behavior reports, and threat intelligence without executing the malware locally.



The taken sample hash - 44d88612fea8a8f36de82e1278abb02f

Detection ratio: 64 out of 67[illegible]

VirusTotal - File - 275a021bbf6489e544471899f7db9d1663fc95ec2fe2a2c4538aab651fd0t-relations

275a021bbf6489e544471899f7db9d1663fc95ec2fe2a2c4538aab651fd0t-relations

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

Join our Community and enjoy additional community insights and crowd-sourced detections, plus an API key to automate checks.

Contacted Domains (19)

Domain	Detections	Created	Registrar
alibaba-inc.com	0 / 92	1999-03-08	MarkMonitor Inc.
aliexpress.com	0 / 92	1999-03-02	MarkMonitor Inc.
apple-cloudkit.apple-dns.net	0 / 92	2014-05-28	-
api.aspectrf.in	1 / 92	2016-05-16	MarkMonitor Inc.
apple.monastic.com	0 / 92	2010-07-17	-
asurion-fido.com	0 / 92	1996-11-17	-
awscli.amazonaws.com	0 / 92	1994-11-11	MarkMonitor Inc.
awscli.amazonaws.com.ses.amazonaws.com	0 / 92	2002-04-02	MarkMonitor Inc.
bq.amazonaws.com.fds.amazonaws.com	0 / 92	2011-04-18	MarkMonitor Inc.
cgis0007025.gdn.co	1 / 92	2014-06-11	Safenames Ltd

Contacted IP addresses (110)

IP	Detections	Autonomous System	Country
100.22.10.188	0 / 92	1609	US
104.08.61.118	0 / 92	20940	US
104.88.206.137	0 / 92	20940	US
104.88.206.141	0 / 92	20940	US
104.88.206.142	0 / 92	20940	US
104.88.206.143	0 / 92	20940	US
104.88.206.145	0 / 92	20940	US
104.88.206.146	0 / 92	20940	US
104.88.206.147	0 / 92	20940	US
104.88.206.149	0 / 92	20940	US

VirusTotal - File - 275a021b0bf6489e54d471899f7db9d1663f6c69sec2f62a2c4538aabf651fd0f/relations

Execution Parents (3.4 K)

Scanned	Detections	Type	Name
2025-09-28	0 / 62	Shell script	linux.sh
2025-12-19	10 / 34	ZIP	file2ec023056d011686c54a500fa07af59
2025-10-31	37 / 64	PDF	pdf-doc-vba-eicar-dropper.pdf
2025-01-09	46 / 72	Win32 EXE	我应病毒.exe
2025-05-25	55 / 65	ZIP	augustus4.18-deobfuscated.zip
2025-12-16	21 / 44	Android	metasploit-framework.zip
2023-03-18	35 / 58	ZIP	01283517142600e09a0638180462487285a17c1f32b28bde6b7d1d2b07625a09
2026-01-21	0 / 61	JavaScript	app.js
2025-05-16	1 / 60	Shell script	linux.sh
2025-12-17	21 / 44	Android	file2ec023056d011686c54a500fa07af59

PE Resource Parents (100)

Scanned	Detections	Type	Name
2025-04-26	46 / 71	Win32 EXE	test09.exe
2025-01-09	46 / 72	Win32 EXE	我应病毒.exe
2025-03-03	41 / 72	Win32 EXE	9201955a7b9dfc718a4342c41558bd4
2023-13-27	16 / 71	Win32 EXE	Test2.exe
2025-10-19	64 / 72	Win32 EXE	bio223a.exe
2024-01-31	33 / 71	Win32 EXE	MMT
2023-05-20	43 / 67	Win32 EXE	神威之燃.exe
2025-09-25	10 / 71	Win32 EXE	Vb6NoOpt
2025-13-22	32 / 72	Win32 EXE	sample.exe
2024-03-02	50 / 72	Win32 EXE	Kmq5ACvRCBh.exe

Start

VirusTotal - File - 275a021b0bf6489e54d471899f7db9d1663f6c69sec2f62a2c4538aabf651fd0f/relations

275a021b0bf6489e54d471899f7db9d1663f6c69sec2f62a2c4538aabf651fd0f

Dropped Files (83)

Scanned	Detections	File type	Name
2025-11-09	0 / 64	Mach-O	JCT_rv18da
2025-11-09	0 / 64	Mach-O	JCT_byeGeF
2020-07-27	0 / 61	Mach-O	JCT_Bt0Ac
2020-07-04	0 / 62	Mach-O	JCT_P0ym0
2020-07-04	0 / 61	Mach-O	JCT_H0M0tp
2023-05-03	0 / 60	Mach-O	JCT_y0M4d
2021-11-00	0 / 58	Mach-O	JCT_rglR0R
2024-01-20	0 / 61	XML	RemoteConfiguration.plist
2021-11-02	0 / 59	Mach-O	JCT_J0d4H
2026-01-21	0 / 61	Text	2022.txt

Graph Summary

- Analyze detection reports:

Malware name: DOS.EiracA.Trojan

Threat category: EICAR test file

Behavior analysis: Malware trojan evader

Network activity: 32 DNS, 24 IP and many more are identified while executing this file.

Crowdsourced Sigma Rules

CRITICAL 0HIGH 0MEDIUM 0LOW 1

Matches rule **User with Privileges Logon** by frack113 at Sigma Integrated Rule Set (GitHub)

↳ Detects logon with "Special groups" and "Special Privileges" can be thought of as Administrator groups or privileges.

Crowdsourced IDS rules

HIGH 0MEDIUM 0LOW 1INFO 0

Matches rule **(tcp) experimental TCP options found** at Snort registered user ruleset

↳ protocol-command-decode

Behavior Tags

checks-cpu-name detect-debug-environment direct-cpu-clock-access idle long-sleeps sets-process-name

Dynamic Analysis Sandbox Detections

The sandbox Zenbox flags this file as: MALWARE TROJAN

Files Dropped

+ 590aee7bdd69b59b.customDestinations-ms

+ 590aee7bdd69b59b.customDestinations-ms~RFdc94.TMP

+ StartupProfileData-NonInteractive

+ __PSScriptPolicyTest_sxp1lrt4.hj2.psm1

+ powershell.exe.log

+ /private/var/folders/8s/wczf490s3zxb_m1q9d3sw90r0000gn/T/com.apple.remindd/TemporaryItems/NSIRD_remindd_he99qB/RemoteConfiguration.plist

+ C:\Users\<USER>\AppData\Local\Microsoft\Windows\PowerShell\StartupProfileData-NonInteractive

+ C:\Users\<USER>\AppData\Local\Temp\OutofProcReport239420.txt

+ C:\Users\<USER>\AppData\Local\Temp\pm5ljsiy.xo4.ps1

+ /proc/5325/coredump_filter

VirusTotal - File - 275a021bbfb...

virustotal.com/gui/file/275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aab6f651f0f/behavior

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aab6f651f0f

etcar.com 26142

Size 68 B

Last Analysis Date 18 minutes ago

Community Score 3714

powerhell idle attachment known-distributor long-sleeps detect-debug-environment direct-cpu-clock-access via-tor

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Display grouped sandbox reports

CAPE Linux

1

1

0

0

0

4

CAPE Sandbox

5

0

0

0

5

0

Lastline

2

0

0

0

0

0

0

OS X Sandbox

3

6

0

0

0

1

39

VirusTotal Jupyterbox

0

0

0

0

0

0

0

VirusTotal Observer

0

0

0

0

1

0

0

Yara Hunter

0

1

0

0

0

0

0

Zenbox

2

6

0

0

1

80

19

Activity Summary

Download Artifacts

Full Reports

Help

3 Detections

3 MALWARE

3 TROJAN

1 EVASION

4 Mitre Signatures

1 LOW

0 MEDIUM

0 IDS Rules

1 LOW

0 Sigma Rules

1 LOW

5 Dropped Files

7 OTHER

1 TEXT

1 IMAGE

6 Network comms

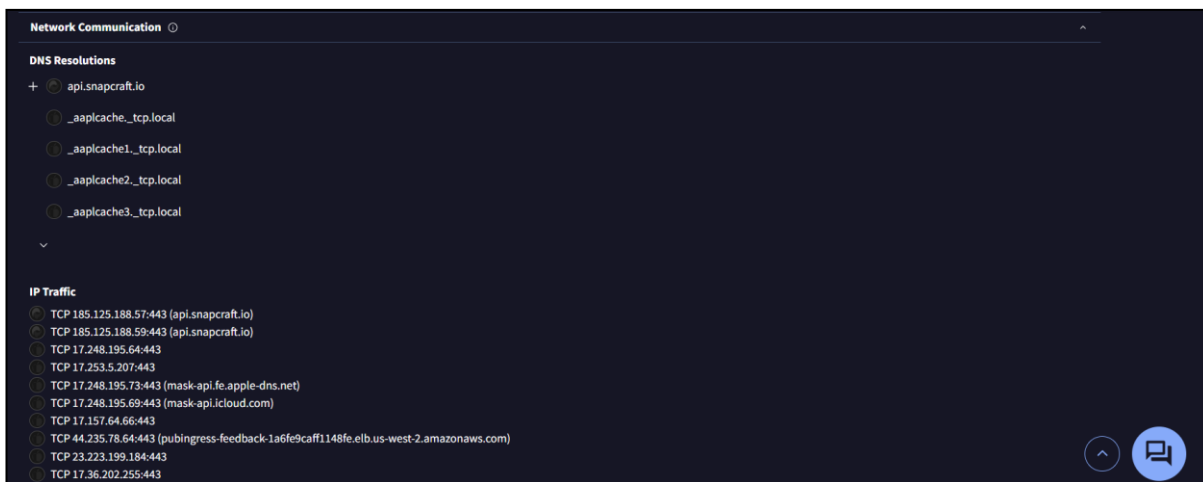
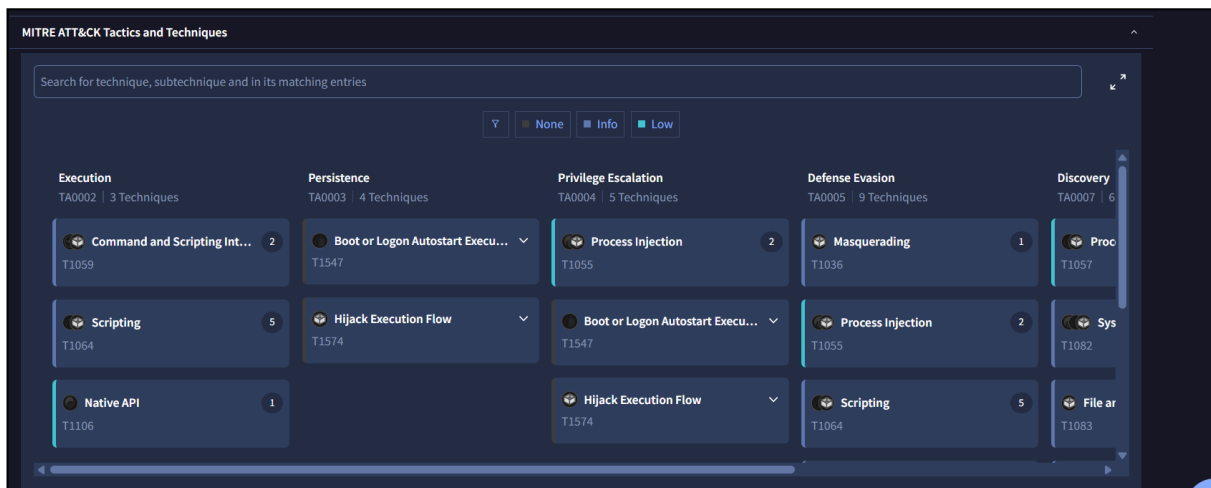
35 DNS

34 IP

2 JAR

Behavior Tags

checks-cpu-name detect-debug-environment direct-cpu-clock-access idle long-sleeps sets-process-name



Registry findings

- **Observe behavior indicators:**

The observed behavior indicators show that the malware attempts to **modify system files and registry entries**, which suggests persistence mechanisms. By making registry changes, the malware ensures it can automatically execute after system reboot. The malware also demonstrates **file system**

activity, such as creating or dropping additional files in system directories. This behavior indicates that the malware may install secondary payloads or supporting components required for its operation.

These are the major behavior analysis observed after analysing that hash in the irustotal application.

- **Malware lifecycle:**

Creation/Development:

Malware is designed by attackers to achieve specific goals, such as stealing data, disrupting systems, or encrypting files. Techniques like obfuscation, packing, and polymorphism help it avoid detection.

Delivery/Distribution:

Malware is delivered to the target through phishing emails, malicious downloads, infected USBs, fake software updates, or exploiting system vulnerabilities.

Exploitation:

Once delivered, malware takes advantage of system flaws or user actions to execute its code.

Installation/Persistence:

Malware installs itself and ensures it stays on the system, often hiding from the user or antivirus software, using methods like registry modification, services, or rootkits.

Command & Control (C2):

Some malware connects to remote servers controlled by attackers to receive commands, exfiltrate data, or update itself.

Actions on Objectives:

Malware carries out its intended purpose, such as stealing sensitive information, encrypting files (ransomware), disrupting systems, or spreading to other devices.

Detection/Removal:

Eventually, malware may be detected and removed by antivirus tools or system scans, though some damage, like stolen data or system changes, may remain.

- **How malware spreads:**

Malware spreads using various techniques depending on its type.

Common spreading methods include:

- ✓ Phishing emails with malicious attachments or links
- ✓ Drive-by downloads from compromised websites
- ✓ Exploiting unpatched system vulnerabilities
- ✓ Infected USB drives or external storage
- ✓ Pirated or cracked software downloads
- ✓ Network-based spreading using open ports and services

Modern malware often uses social engineering to trick users rather than relying only on technical vulnerabilities.

- **Prevention methods:**

Preventing malware requires a combination of technical controls and user awareness.

Key prevention methods include:

- ✓ Installing and updating antivirus and anti-malware software
- ✓ Regularly patching operating systems and applications
- ✓ Avoiding suspicious links and unknown email attachments
- ✓ Using strong passwords and multi-factor authentication
- ✓ Disabling macros in documents from unknown sources
- ✓ Regular data backups to mitigate ransomware attacks
- ✓ Using firewalls and intrusion detection systems

User education plays a vital role in preventing malware infections.

- **Summary:**

Using VirusTotal for malware analysis helps in understanding how malware is detected and classified. It shows how dangerous a file is by checking its detection rate and behavior. Learning about different types of malware, how they spread, and how they work makes it easier to identify threats early. Behavior-based analysis, along with signature-based detection, improves accuracy in finding malware. This task helps build a basic understanding of malware and how to detect it in real-world situations.